

DAFTAR LAMBANG DAN SINGKATAN

SINGKATAN	NAMA
AI	<i>Artificial Intelligence</i> (Kecerdasan Buatan)
ANRI	Arsip Nasional Republik Indonesia
API	<i>Application Programming Interface</i>
BCE	<i>Binary Cross-Entropy</i>
BiGRU	<i>Bidirectional Gated Recurrent Unit</i>
BiLSTM	<i>Bidirectional Long Short-Term Memory</i>
CBAM	<i>Convolutional Block Attention Module</i>
CER	<i>Character Error Rate</i> (Tingkat Kesalahan Karakter)
cGAN	<i>Conditional Generative Adversarial Network</i>
CLI	<i>Command-Line Interface</i>
CNN	<i>Convolutional Neural Network</i> (Jaringan Saraf Konvolusional)
CRNN	<i>Convolutional Recurrent Neural Network</i>
CTC	<i>Connectionist Temporal Classification</i>
CUDA	<i>Compute Unified Device Architecture</i>
cuDNN	<i>CUDA Deep Neural Network library</i>
CWV	<i>Core Web Vitals</i>
DE-GAN	<i>Document Enhancement GAN</i>
DIBCO	<i>Document Image Binarization Contest</i>
DocEnTr	<i>Document Enhancement Transformer</i>
DRD	<i>Distance Reciprocal Distortion</i>
DSRM	<i>Design Science Research Methodology</i>
ERB	<i>Enhance to Read Better</i>
FCN	<i>Fully Convolutional Network</i>
FM	<i>F-measure</i>
Fps	<i>pseudo-F-measure</i>
FR	<i>Functional Requirement</i> (Kebutuhan Fungsional)
GAN	<i>Generative Adversarial Network</i> (Jaringan Adversarial Generatif)
GPU	<i>Graphics Processing Unit</i>
GT	<i>Ground Truth</i> (Kebenaran Dasar)
H-DIBCO	<i>Handwritten Document Image Binarization Contest</i>
HMM	<i>Hidden Markov Model</i>
HTR	<i>Handwritten Text Recognition</i> (Pengenalan Teks Tulisan Tangan)

IAM	<i>IAM Handwriting Database</i>
ICDAR	<i>International Conference on Document Analysis and Recognition</i>
IIIT5K	<i>IIIT 5K-word dataset</i>
KHATT	<i>Arabic Handwritten Text Database</i>
LSTM	<i>Long Short-Term Memory</i>
MAE	<i>Mean Absolute Error</i> (Galat Absolut Rata-Rata)
ML	<i>Machine Learning</i> (Pembelajaran Mesin)
MLflow	<i>Machine Learning Flow</i>
MSE	<i>Mean Squared Error</i> (Galat Kuadrat Rata-Rata)
MSFP	<i>Multi-Scale Feature Pyramid</i>
NFR	<i>Non-Functional Requirement</i> (Kebutuhan Non-Fungsional)
OCR	<i>Optical Character Recognition</i> (Pengenal Karakter Optik)
PALM	<i>Presentation Attack Liveness Detection in Mobile scenarios</i>
Pix2Pix	<i>Image-to-Image Translation with Conditional Adversarial Networks</i>
PSNR	<i>Peak Signal-to-Noise Ratio</i> (Rasio Puncak Sinyal terhadap Derau)
RDB	<i>Residual Dense Block</i>
RMSProp	<i>Root Mean Square Propagation</i>
RNN	<i>Recurrent Neural Network</i> (Jaringan Saraf Berulang)
SDM	Sumber Daya Manusia
SGD	<i>Stochastic Gradient Descent</i>
SNR	<i>Signal-to-Noise Ratio</i> (Rasio Sinyal terhadap Derau)
SOP	<i>Standard Operating Procedure</i> (Prosedur Operasi Standar)
SOTA	<i>State-of-the-Art</i>
SSIM	<i>Structural Similarity Index Measure</i> (Indeks Kesamaan Struktural)
UNESCO	<i>United Nations Educational, Scientific and Cultural Organization</i>
U-Net	<i>U-shaped Convolutional Network</i>
VGG	<i>Visual Geometry Group</i>
ViT	<i>Vision Transformer</i>
VOC	<i>Visual Object Classes</i>
WER	<i>Word Error Rate</i> (Tingkat Kesalahan Kata)

LAMBANG

NAMA

Huruf Yunani (Parameter Model)

α	Koefisien bobot untuk <i>loss function</i> , tingkat signifikansi statistik
β	Koefisien bobot untuk <i>Binary Cross-Entropy loss</i>
λ	Koefisien bobot untuk <i>CTC loss</i> , parameter regularisasi
θ	Parameter model <i>neural network</i>
σ	Deviasi standar
μ	Rata-rata (<i>mean</i>)
ε	<i>Token blank</i> /kosong dalam CTC, konstanta kecil untuk stabilitas numerik
Δ	Delta (selisih/perubahan nilai)
π	<i>Alignment path</i> dalam CTC
π_t	<i>Token</i> pada <i>timestep</i> t dalam <i>alignment path</i>

Operator Matematis

∇ atau ∂	Operator gradien atau turunan parsial
Σ	Operator penjumlahan
Π	Operator perkalian
$\int$	Operator integral
argmax	Argumen yang memaksimalkan fungsi
$\sim$	Terdistribusi menurut
$\rightarrow$	Menuju atau konvergen ke

Fungsi Loss dan Kerugian

$\mathcal{L}$	Fungsi <i>loss</i> (kerugian)
$\mathcal{L}_{total}$	Total <i>loss function</i>
$\mathcal{L}_{adversarial}$ atau $\mathcal{L}_{adv}$	<i>Adversarial loss</i>
$\mathcal{L}_{reconstruction}$	<i>Reconstruction loss</i>
$\mathcal{L}_{L1}$	L1 <i>loss</i> (<i>Mean Absolute Error</i>)
$\mathcal{L}_{L2}$	L2 <i>loss</i> (<i>Mean Squared Error</i>)
$\mathcal{L}_{CTC}$	CTC (<i>Connectionist Temporal Classification</i>) <i>loss</i>
$\mathcal{L}_{perceptual}$	<i>Perceptual loss</i>
$\mathcal{L}_{pixel}$	<i>Pixel-wise loss</i>
$\mathcal{L}_{BCE}$	<i>Binary Cross-Entropy loss</i>
$\mathcal{L}_{GAN}$	GAN <i>loss</i>
$\mathcal{L}_{cycle}$	<i>Cycle consistency loss</i>

Variabel Model dan Komponen GAN

G	Generator dalam GAN
D	Discriminator dalam GAN
$G(z)$	Output dari generator dengan input z
$D(x)$	Output dari discriminator dengan input x
$G_{A \rightarrow B}$	Generator dari domain A ke B
$G_{B \rightarrow A}$	Generator dari domain B ke A
$V(D, G)$	Fungsi nilai (<i>value function</i>) dalam GAN
x	Sampel data asli, <i>input</i> citra
y	Label atau <i>ground truth</i> , kondisi dalam <i>conditional GAN</i>
z	Noise vector atau representasi laten

Distribusi Probabilitas

$\mathbb{E}$	Ekspektasi atau nilai harapan
p_{data}	Distribusi data asli
p_z	Distribusi <i>prior</i> (<i>noise</i>)
$p(y x)$	Probabilitas bersyarat <i>output</i> y given <i>input</i> x
$p_t(\pi_t x)$	Probabilitas <i>token</i> pada <i>timestep</i> t

Fungsi dan Transformasi CTC

$\mathcal{B}$	Fungsi penciutan (<i>collapse function</i>) dalam CTC
$\mathcal{B}^{-1}$	Fungsi invers penciutan dalam CTC
T	Jumlah <i>timestep</i> atau panjang <i>sequence</i>

Metrik Jarak dan Norma

$ x - G(y) _1$	L1 <i>distance</i> antara <i>ground truth</i> dan <i>generated image</i>
$ x - G(y) _2$	L2 <i>distance</i> antara <i>ground truth</i> dan <i>generated image</i>
$N \times N$	Ukuran <i>patch</i> dalam PatchGAN

Statistik dan Analisis

d	Cohen's d (<i>effect size</i>)
n	Jumlah sampel
p	p -value (nilai probabilitas statistik)
r	Koefisien korelasi
R^2	Koefisien determinasi

Notasi Kompleksitas Komputasi

$O(n)$	Notasi Big-O untuk kompleksitas linear
$O(n^2)$	Notasi Big-O untuk kompleksitas kuadratik